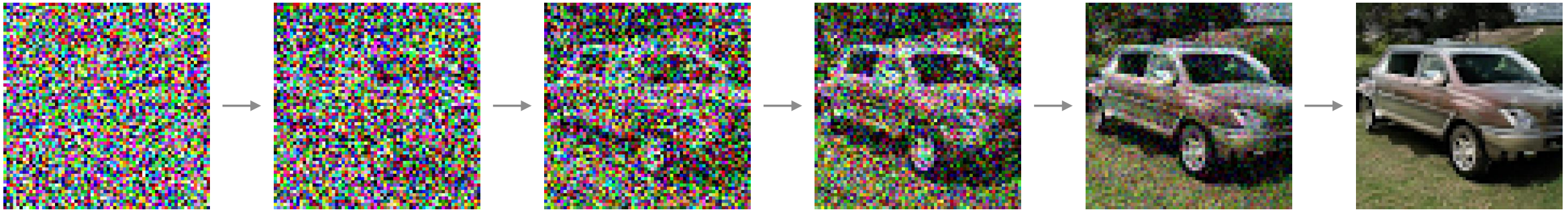
Improved Denoising Diffusion Probabilistic Models

Alex Nichol and Prafulla Dhariwal, ICML 2021

Presented by Neric Joel A

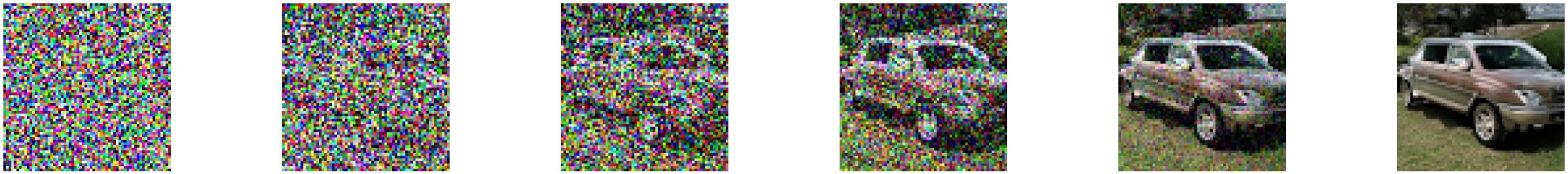
15 October 2026



Generating one image with the paper's released ImageNet 64×64 model on my laptop (6 of its 250 steps shown).

DDPM in 2020: good samples, weak likelihood, slow sampling

Forward process: add a little Gaussian noise at each step (fixed, nothing is learned)



Reverse process: a network removes the noise, one network call per step (learned)

Weak likelihood

3.70 bits per dimension on CIFAR-10, against **3.14** for PixelCNN (lower is better).

Likelihood is read as how well a model covers its data.

Slow sampling

One network call per step. Trained with T = 4000 steps, one image takes several minutes on a GPU.

Every call costs latency and energy, paid again for each image served.

Numbers: Nichol & Dhariwal (2021), Table 3 and Sec. 4. Images: my run of the released ImageNet 64×64 checkpoint (github.com/openai/improved-diffusion).

Three changes: learned variance, cosine schedule, fewer steps

1 Learn the variance of each reverse step

$$\Sigma_\theta(x_t, t) = \exp\left(v \log \beta_t + (1 - v) \log \tilde{\beta}_t\right)$$

$$L_{\text{hybrid}} = L_{\text{simple}} + \lambda L_{\text{vlb}}, \quad \lambda = 0.001$$

The original DDPM fixed it; a small extra loss term trains it.

2 Cosine noise schedule

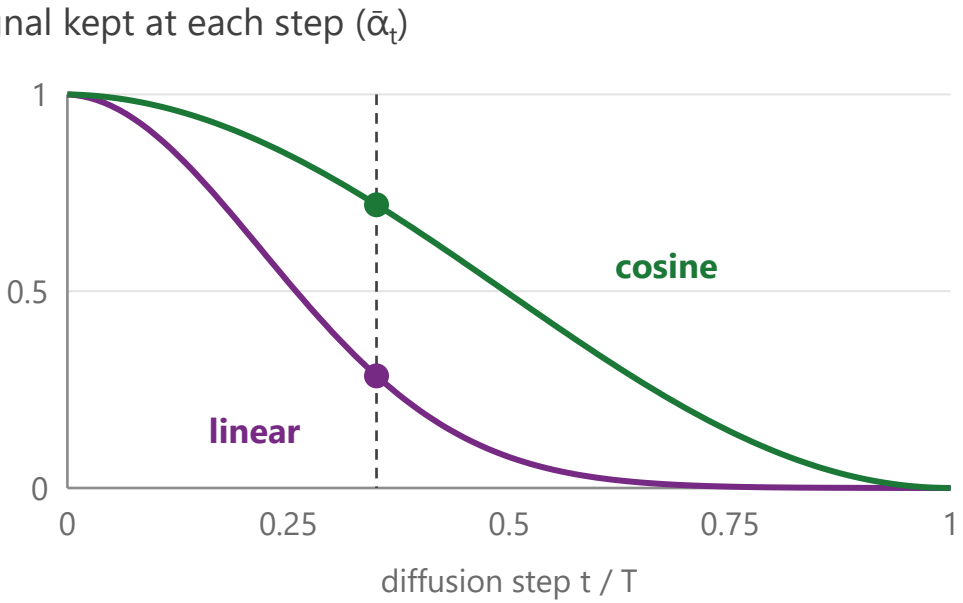
$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad f(t) = \cos^2\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right)$$

Destroys information more slowly than the linear schedule (right).

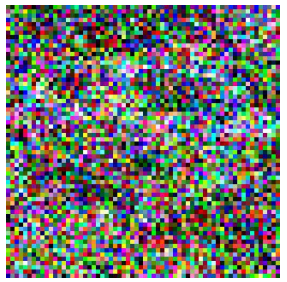
3 Sample with fewer steps

$$\beta_{S_t} = 1 - \frac{\bar{\alpha}_{S_t}}{\bar{\alpha}_{S_{t-1}}}, \quad K \ll T \text{ steps}$$

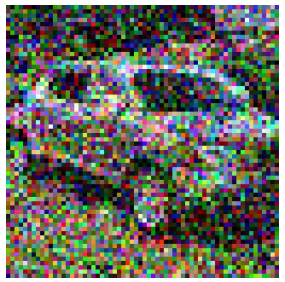
Respace β to K steps; the learned variance rescales itself.



linear, t/T = 0.35



cosine, t/T = 0.35



Steps that are almost pure noise

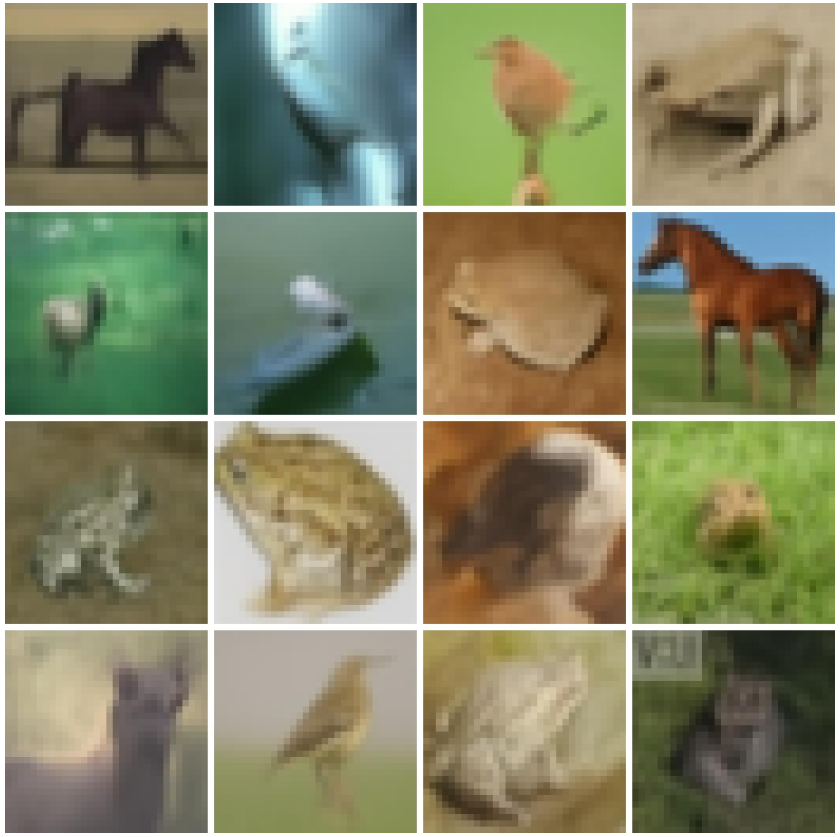
linear **23%**

cosine **3%**

Eq. 15 to 17 and 19: Nichol & Dhariwal (2021). Curves and images: my reproduction of their Fig. 3 and 5 (same picture and noise for both).

At 10 steps, only the learned variance gives clean images

Learned Σ_θ this paper



Fixed $\sigma^2 = \beta_t$ original DDPM



Released CIFAR-10 checkpoint, running on my laptop.

Trained with 4000 steps, sampled with 10. No retraining.

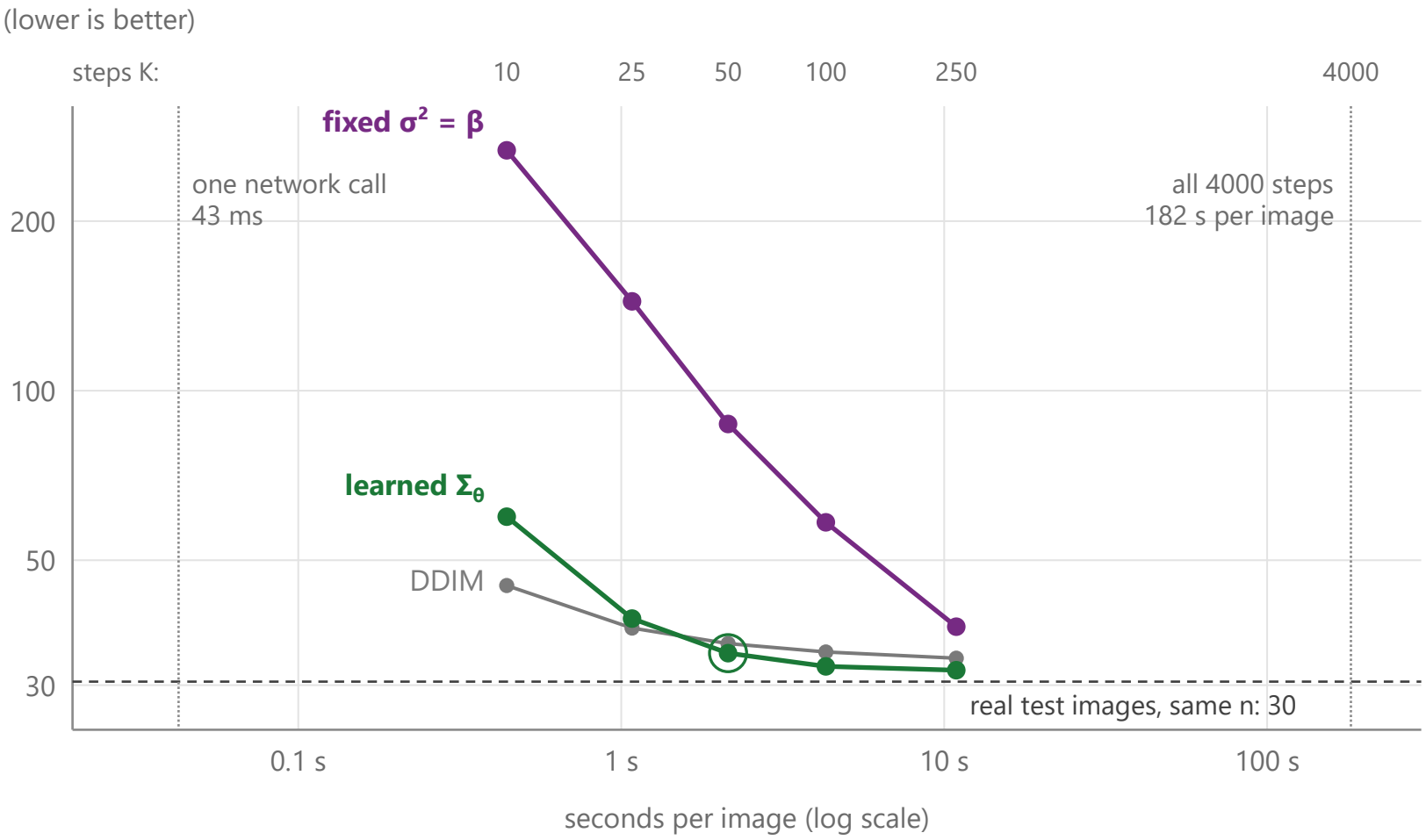
Same network and same starting noise in both grids. Only the variance rule differs.

FID on 1,000 samples: **59.7** learned, **267.2** fixed.

Live demo

My run of the released CIFAR-10 L_hybrid checkpoint on an RTX 3050 Ti laptop GPU. Compare the paper's Fig. 8.

Quality holds at 50 steps, still 50 network calls per image



At 50 steps (circled): FID **34** learned, **87** fixed. 2.1 s per image, 85× faster than all 4000 steps.

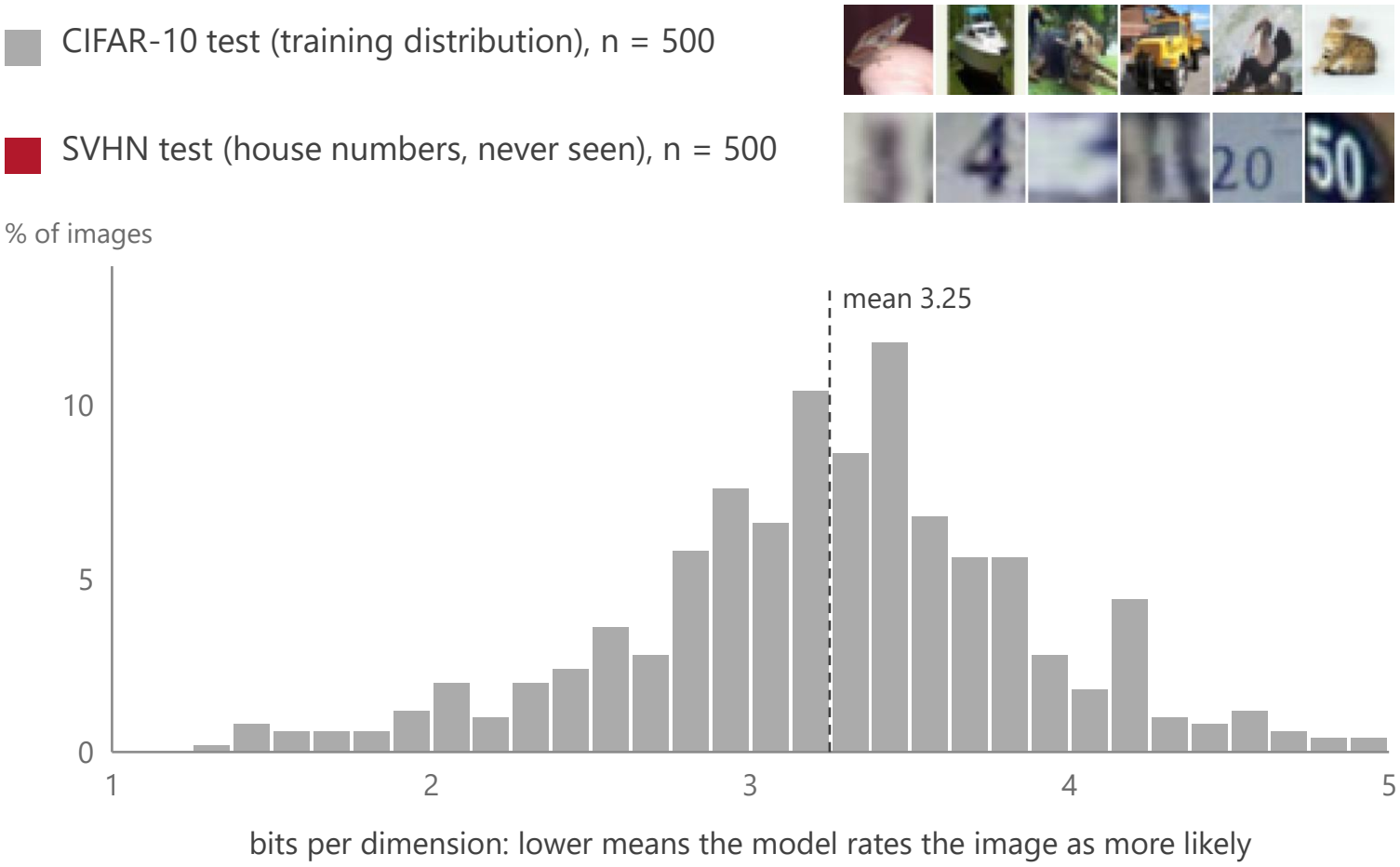
A GAN makes an image in one network call.

DDIM is better at 10 and 25 steps, as the paper also reports.

The paper reports steps, never seconds or energy per image.

My measurements on the released CIFAR-10 checkpoint: FID on 1,000 samples (so higher than the paper's 50,000-sample values), time per image at batch size 1.
DDIM: Song et al. (2021).

Train on CIFAR-10, test on SVHN: does the likelihood notice?



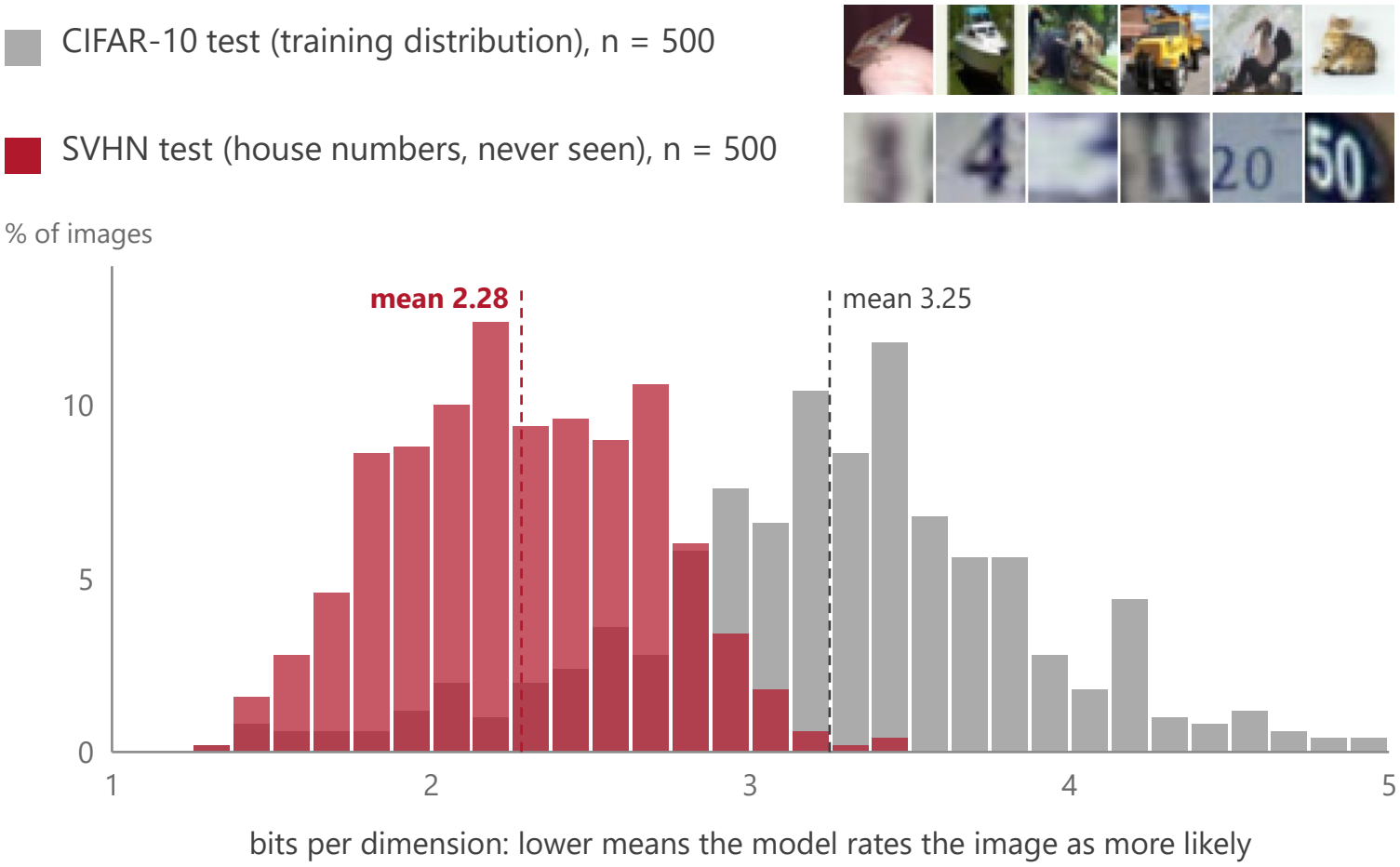
Lower bits per dimension means the model rates the image as more likely.

If likelihood reflects what the model has learned, images it has never seen should land to the right.

Where do the SVHN digits land?

My run: NLL bound with the paper's Appendix E schedule, released CIFAR-10 checkpoint, 500 test images per set. CIFAR-10: Krizhevsky (2009). SVHN: Netzer et al. (2011). Compare Heng et al. (2024); Nalisnick et al. (2019).

The model finds unseen SVHN digits more likely than CIFAR-10



Used as an alarm for unfamiliar inputs

0.10 AUROC

0.5 is chance and 1.0 a perfect alarm, so the ranking is backwards.

Input	AUROC	
Heavy pixel noise	**1.00**	caught
Upside-down photos	**0.53**	chance
SVHN digits	**0.10**	backwards

A plain black image scores 0.11 bits/dim, more likely than any photo.

Same failure as Glow and PixelCNN (Nalisnick et al. 2019). Heng et al. (2024) report 0.091 on this checkpoint.

My run: NLL bound with the paper's Appendix E schedule, released CIFAR-10 checkpoint, 500 test images per set. CIFAR-10: Krizhevsky (2009). SVHN: Netzer et al. (2011). Compare Heng et al. (2024); Nalisnick et al. (2019).

Train on one, test on another: revealing, but setup-sensitive

Pros	Cons
Exposes what in-distribution numbers hide. The likelihood looked fine on CIFAR-10 and failed on SVHN.	**Direction matters.** Trained on SVHN and tested on CIFAR-10, the likelihood separates them well.
Cheap. Released checkpoint, no new labels, about 10 minutes per dataset pair on a laptop GPU.	**The score matters.** A different likelihood-based score reaches 0.94 on the same pair.
Resembles deployment. Real inputs drift away from the training data.	**My number is a bound.** A 140-step NLL bound on 500 images per set: fine for ranking, not an exact NLL.

In practice: fix the dataset pairs in advance, test both directions, and report more than one score.

Heng et al. (2024): trained on SVHN, tested on CIFAR-10, AUROC 0.99. Goodier & Campbell (2023): 0.94 on CIFAR-10 versus SVHN with a different DDPM and score.

The speed claim holds; the likelihood claims hold only in part

Claim in the paper	Evidence	Verdict
50 to 100 sampling steps with little loss in quality	Fig. 8. My run at 50 steps: FID **34** with the learned variance, **87** with the fixed one; real images 30.	**Supported** but 2.1 s and 50 network calls per image
Competitive likelihood while keeping sample quality	Tables 2 and 3: the best NLL, **2.94**, comes from L_{vlb} (FID 11.5). The good-sample model, L_{hybrid}, has NLL **3.17**, behind PixelCNN's 3.14.	**Partly** two different models
Implied: better likelihood means the model knows the data	Trained on CIFAR-10, tested on SVHN: AUROC **0.10** in my run, 0.091 in Heng et al. (2024).	**Fails under shift** do not use it to flag unusual inputs

Nichol & Dhariwal (2021), Tables 2 and 3, Fig. 8; my runs on the released checkpoints; Heng et al. (2024).

Before deployment, I would measure more than FID and NLL

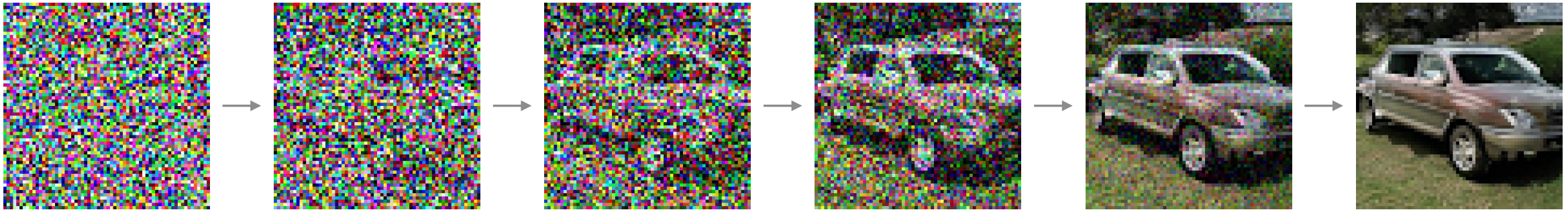
Measure	Why, for this model	How
Shift detection	Likelihood rates SVHN as more typical than CIFAR-10 (AUROC 0.10).	OOD AUROC in both directions, on several dataset pairs.
Memorization	301 training images were extracted from 1M samples of this checkpoint (Carlini et al.).	Extraction rate; membership inference at a low false-positive rate.
Cost per image	The paper reports steps, never seconds or energy.	Latency and kWh per 1,000 images; plot FID against cost.
Coverage by class and group	One recall number can hide classes or groups the model drops.	Per-class recall; train a classifier on samples, test it on real images.
Provenance	Detectors trained on GAN images do worse on diffusion images (Ricker et al.).	Detector AUROC on the model's samples; watermark robustness.

Carlini et al. (2023); Luccioni, Jernite & Strubell (2024); Ravuri & Vinyals (2019); Ricker et al. (2024).

Thank you

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15 October 2026



Use of generative AI. Claude (Anthropic) helped me study the paper, write the experiment and demo code, and draft these slides. ChatGPT and Codex (OpenAI) gave feedback on the demo design. Every number comes from the paper or from my own runs of the released checkpoints; sources are cited on each slide.